

Analysis

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Analysis

Packages

```
set.seed(157)
library(tidyverse)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.0.4
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

Functions Script

```
setwd('/Users/camsmithers/Desktop/Camalytics/NBA')
source('inR/Functions.R')
```

Data Sets

Loading in the data sets that I'll need for analysis.

```
setwd('/Users/camsmithers/Desktop/Camalytics/NBA')
team_box_scores <- readRDS(
  'Data-NBA/Current/team_box_scores_updated.rds')
player_box_scores <- readRDS(
  'Data-NBA/Current/full_box_player_updated.rds')
team_season_stats <- readRDS(
  'Data-NBA/Current/team_season_stats_updated.rds')
player_season_stats <- readRDS(
  'Data-NBA/Current/player_season_stats_updated.rds')
```

Applying a function I wrote that fixes some aesthetic issues with player names. This will occur with foreign players that have special characters or foreign characters in their name.

```
player_box_scores <- fixplayername(player_box_scores)
player_season_stats <- fixplayername(player_season_stats)
```

Outcomes and Small Metrics

Outcome I'm adding the points from the opposing team so I can create an outcome variable based on the margin of the outcome. This will be multinomial regression with four different classifications. Those are...

```
opponent_stats <- team_box_scores %>%
  select(gamedate, team, pts) %>%
  rename('opp_pts'='pts')

team_box_scores <- team_box_scores %>%
  left_join(opponent_stats, by = c('gamedate', 'opponent'='team'))

team_box_scores <- team_box_scores %>%
  mutate(
    ptsdiff = pts - opp_pts,
    twoleveloc = if_else(offrating > defrating, 1, 0),
    threeleveloc = case_when(
      ptsdiff >= 11 ~ 2,
      ptsdiff %in% 1:10 ~ 1,
      TRUE ~ 0),
    fourleveloc = case_when(
      ptsdiff <= -11 ~ 0,
      ptsdiff %in% -1:-10 ~ 1,
      ptsdiff %in% 1:10 ~ 2,
      ptsdiff >= 11 ~ 3)) %>%
  mutate(
    twoleveloc = factor(twoleveloc,
      levels = c(0, 1),
      labels = c('Loss', 'Win')),
    threeleveloc = factor(threeleveloc,
      levels = c(0, 1, 2),
      labels = c('Loss', 'Basic Win', 'Blowout Win')),
    fourleveloc = factor(fourleveloc,
      levels = c(0, 1, 2, 3),
      labels = c('Blowout Loss', 'Basic Loss',
        'Basic Win', 'Blowout Win')))
```

Conditional Outcomes I want to look at how missing a player affects the chance of a team to win a game; however, I have to determine which players are important for the outcome of the game. I'm going to look at few metrics and use those to determine a player's significance. For the different metrics, different players may end up being the most important player. Those metrics include...

1. Win Shares: `ws`
2. Value Over Replacement Player (VORP): `vorp`
3. Player Efficiency Rating: `player_effrtg`

I'm also going to create a variable that has a unique id for each best player by a given metric.

```
bp_by_ws <- best_player_metric(player_season_stats, ws, 20, 10) %>%
  rename('ws_name'='name') %>%
  unite('ws_team_id', c('year', 'team', 'ws_player_id'),
    sep = '_', remove = FALSE) %>%
  mutate(ws_team_id = paste0('ws_', ws_team_id)) %>%
  relocate(ws_team_id, .after = last_col())
bp_by_vorp <- best_player_metric(player_season_stats, vorp, 20, 10) %>%
  rename('vorp_name'='name') %>%
```

```

unite('vorp_team_id', c('year', 'team', 'vorp_player_id'),
      sep = '_', remove = FALSE) %>%
mutate(vorp_team_id = paste0('vorp_', vorp_team_id)) %>%
relocate(vorp_team_id, .after = last_col())
bp_by_per <- best_player_metric(player_season_stats, player_effrtg, 20, 10) %>%
rename('per_name'='name') %>%
unite('per_team_id', c('year', 'team', 'player_effrtg_player_id'),
      sep = '_', remove = FALSE) %>%
mutate(per_team_id = paste0('per_', per_team_id)) %>%
relocate(per_team_id, .after = last_col())

```

Here I'm going to create a variable within each data set that will allow to check for player availability when coding a future outcome variable

Joining the data from the best player by a certain metric data sets to later determine if player was on the court and how that affected the outcome.

```

team_box_scores_2 <- team_box_scores %>%
  left_join(bp_by_ws, by = c('team', 'season'='year')) %>%
  left_join(bp_by_vorp, by = c('team', 'season'='year')) %>%
  left_join(bp_by_per, by = c('team', 'season'='year')) %>%
  mutate(
    ws_tm_game_id = paste0(ws_team_id, '_', gamedate),
    vorp_tm_game_id = paste0(vorp_team_id, '_', gamedate),
    per_tm_game_id = paste0(per_team_id, '_', gamedate))

player_box_scores_2 <- player_box_scores %>%
  left_join(bp_by_ws, by = c('team', 'season'='year', 'name'='ws_name')) %>%
  left_join(bp_by_vorp, by = c('team', 'season'='year', 'name'='vorp_name')) %>%
  left_join(bp_by_per, by = c('team', 'season'='year', 'name'='per_name')) %>%
  mutate(
    ws_plyr_game_id = if_else(
      !is.na(ws_team_id), paste0(ws_team_id, '_', gamedate), NA),
    vorp_plyr_game_id = if_else(
      !is.na(vorp_team_id), paste0(vorp_team_id, '_', gamedate), NA),
    per_plyr_game_id = if_else(
      !is.na(per_team_id), paste0(per_team_id, '_', gamedate), NA))

```

After creating the data sets necessary to determine whether a team's best player (by a given metric) was available I can now create an outcome variable to check for it. I'm also going to add a variable to determines whether a game was a regular season game or playoff game.

Note: *I know I don't have to use multiple mutate functions; however, at times I like to use multiple as it helps me organize my code as I'm writing it.*

```

team_box_scores_3 <- team_box_scores_2 %>%
  mutate(
    ws_bpa = if_else(ws_tm_game_id %in%
      player_box_scores_2$ws_plyr_game_id, 1, 0),
    vorp_bpa = if_else(vorp_tm_game_id %in%
      player_box_scores_2$vorp_plyr_game_id, 1, 0),
    per_bpa = if_else(per_tm_game_id %in%
      player_box_scores_2$per_plyr_game_id, 1, 0)) %>%
  mutate(
    ws_bpa_multi = case_when(
      twolevelloc == 'Loss' & ws_bpa == 0 ~ 0,

```

```

twoleveloc == 'Loss' & ws_bpa == 1 ~ 1,
twoleveloc == 'Win' & ws_bpa == 0 ~ 2,
twoleveloc == 'Win' & ws_bpa == 1 ~ 3),
vorp_bpa_multi = case_when(
  twoleveloc == 'Loss' & vorp_bpa == 0 ~ 0,
  twoleveloc == 'Loss' & vorp_bpa == 1 ~ 1,
  twoleveloc == 'Win' & vorp_bpa == 0 ~ 2,
  twoleveloc == 'Win' & vorp_bpa == 1 ~ 3),
per_bpa_multi = case_when(
  twoleveloc == 'Loss' & per_bpa == 0 ~ 0,
  twoleveloc == 'Loss' & per_bpa == 1 ~ 1,
  twoleveloc == 'Win' & per_bpa == 0 ~ 2,
  twoleveloc == 'Win' & per_bpa == 1 ~ 3)) %>%
mutate(
  ws_bpa_multi = factor(ws_bpa_multi,
                        levels = c(0, 1, 2, 3),
                        labels = c('Loss Without', 'Loss With',
                                   'Win Without', 'Win With')),
  vorp_bpa_multi = factor(vorp_bpa_multi,
                          levels = c(0, 1, 2, 3),
                          labels = c('Loss Without', 'Loss With',
                                      'Win Without', 'Win With')),
  per_bpa_multi = factor(per_bpa_multi,
                        levels = c(0, 1, 2, 3),
                        labels = c('Loss Without', 'Loss With',
                                   'Win Without', 'Win With')))) %>%
mutate(
  gametype = case_when(
    #2021 Season
    (gamedate >= as.Date('2021-05-22') &
     gamedate <= as.Date('2021-07-20')) |
    #2022 Season
    (gamedate >= as.Date('2022-04-16') &
     gamedate <= as.Date('2022-06-16')) |
    #2023 Season
    (gamedate >= as.Date('2023-04-15') &
     gamedate <= as.Date('2023-06-12')) |
    #2024 Season
    (gamedate >= as.Date('2024-04-20') &
     gamedate <= as.Date('2024-06-17')) |
    #2025 Season
    (gamedate >= as.Date('2025-04-19') &
     gamedate <= as.Date('2025-06-22')) ~ 'Playoff Game',
    TRUE ~ 'Regular Season Game')
)

```

Summary Data

```

team_box_summary <- team_box_scores_3 %>%
  group_by(season, team) %>%
  summarise(
    avg_fg_pct = round(mean(fg_pct, na.rm=TRUE), 3),
    avg_3fg_pct = round(mean(threefg_pct, na.rm=TRUE), 3),

```

```
avg_pts = round(mean(pts, na.rm=TRUE), 1),  
avg_offrating = round(mean(offrating, na.rm=TRUE), 1),  
avg_defrating = round(mean(defrating, na.rm=TRUE), 1))
```

`summarise()` has grouped output by 'season'. You can override using the
`.groups` argument.

Data Export for Visualizations

```
setwd('/Users/camsmithers/Desktop/Camalytics/NBA/Data-NBA/Current')  
saveRDS(team_box_scores_3, file = 'teamvisdata.rds')
```